

Comparative Analysis of Depreciation Methods

Straight-Line vs. Diminishing Balance

Introduction

Depreciation is the systematic allocation of the cost of a tangible asset over its useful life. It represents the allocation of an asset's depreciable amount as the asset is consumed or used over time. Depreciation affects reported expense, profit and the carrying value of assets.

The supplied project guideline specifically requires analysis of Straight-Line and Diminishing Balance methods, practical calculations, a 10-year depreciation schedule, answers to advanced questions and visual presentation. This report follows that analytical structure.

Executive Summary

This project compares two depreciation methods—Straight-Line and Diminishing Balance—using the assumptions specified in the supplied project guideline and Excel model. The analysis uses an asset cost of Rs 4,50,000, additional asset cost of Rs 50,000, total asset price of Rs 5,00,000, scrap value of Rs 50,000 and a useful life of 10 years.

Under Straight-Line, annual depreciation is constant at Rs 45,000. Under Diminishing Balance, the model calculates a depreciation rate of approximately 20.5672%, producing higher depreciation in the early years and progressively lower depreciation thereafter. Year 1 depreciation is approximately Rs 1,02,835.88; Year 2 is Rs 81,685.45; and the Year 4 ending book value is approximately Rs 1,99,053.59.

Key conclusion: Both methods recognize total depreciation of Rs 450,000 over the full 10-year period and end at the Rs 50,000 scrap value. The major difference is the timing of expense recognition.

Scope of the project

The project covers asset-price calculation, annual depreciation, depreciation rates, total depreciation, closing book values and year-on-year Diminishing Balance calculations.

Objectives of the project

- Understand the mechanics and formulas of Straight-Line and Diminishing Balance depreciation.
- Calculate the asset price after incorporating additional asset cost.
- Determine annual Straight-Line depreciation and its percentage.
- Calculate the Diminishing Balance rate and generate the 10-year schedule.
- Compare annual expense, book values and total depreciation.
- Translate the results into practical financial insights.

Input Data and Assumptions

Parameter	Value	Meaning
Asset Cost	4,50,000.00	Initial asset cost
Additional Asset Cost	50,000.00	Additional acquisition cost
Asset Price / Depreciable Cost	5,00,000.00	Asset Cost + Additional Cost
Scrap Value	50,000.00	Expected residual value
Useful Life (Years)	10	Depreciation period

Methodology

Straight-Line Method – The Straight-Line method allocates the depreciable amount equally over the useful life.

Formula: Annual Depreciation = (Asset Price - Scrap Value) ÷ Useful Life

Calculation: (Rs 500,000 – Rs 50,000) ÷ 10 = Rs 45,000 per year

Rate: Rs 45,000 ÷ Rs 500,000 = 9.00% per year

Diminishing Balance Method - Under Diminishing Balance, depreciation is calculated as a fixed percentage of the beginning book value. Because book value declines each year, the absolute depreciation expense also declines.

Formula: Depreciation = Beginning Book Value × Diminishing Balance Rate

Rate used in the project: $1 - (\text{Scrap Value} \div \text{Asset Price})^{(1 \div \text{Useful Life})}$

Calculation: $1 - (50,000 \div 5,00,000)^{(1/10)} = 20.5672\%$

Straight-Line Depreciation Schedule

Year	Beginning Book Value	Year on Year Depreciation Amount	Ending Book Value
1	5,00,000.00	45,000.00	4,55,000.00
2	4,55,000.00	45,000.00	4,10,000.00
3	4,10,000.00	45,000.00	3,65,000.00
4	3,65,000.00	45,000.00	3,20,000.00
5	3,20,000.00	45,000.00	2,75,000.00
6	2,75,000.00	45,000.00	2,30,000.00
7	2,30,000.00	45,000.00	1,85,000.00
8	1,85,000.00	45,000.00	1,40,000.00
9	1,40,000.00	45,000.00	95,000.00
10	95,000.00	45,000.00	50,000.00

Straight-Line produces a constant annual charge of Rs 45,000. The book value falls uniformly from Rs 500,000 to Rs 50,000.

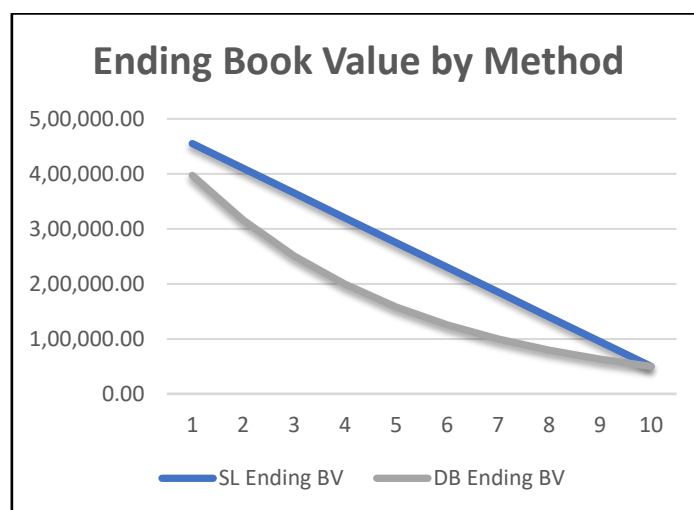
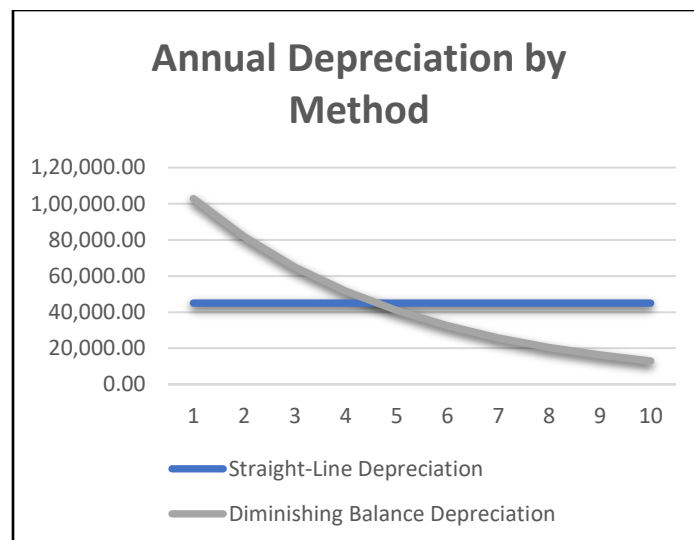
Diminishing Balance Depreciation

Year	Beginning Book Value	Year on Year Depreciation Amount	Ending Book Value
1	5,00,000.00	1,02,835.88	3,97,164.12
2	3,97,164.12	81,685.45	3,15,478.67
3	3,15,478.67	64,885.06	2,50,593.62
4	2,50,593.62	51,540.03	1,99,053.59
5	1,99,053.59	40,939.70	1,58,113.88
6	1,58,113.88	32,519.56	1,25,594.32
7	1,25,594.32	25,831.21	99,763.12
8	99,763.12	20,518.46	79,244.66
9	79,244.66	16,298.39	62,946.27
10	62,946.27	12,946.27	50,000.00

The Diminishing Balance pattern is front-loaded: depreciation is highest in Year 1 and declines each year as the carrying value falls.

Comparative Analysis Metric

Year	Straight-Line Depreciation	Diminishing Balance Depreciation	SL Ending BV	DB Ending BV
1	45,000.00	1,02,835.88	4,55,000.00	3,97,164.12
2	45,000.00	81,685.45	4,10,000.00	3,15,478.67
3	45,000.00	64,885.06	3,65,000.00	2,50,593.62
4	45,000.00	51,540.03	3,20,000.00	1,99,053.59
5	45,000.00	40,939.70	2,75,000.00	1,58,113.88
6	45,000.00	32,519.56	2,30,000.00	1,25,594.32
7	45,000.00	25,831.21	1,85,000.00	99,763.12
8	45,000.00	20,518.46	1,40,000.00	79,244.66
9	45,000.00	16,298.39	95,000.00	62,946.27
10	45,000.00	12,946.27	50,000.00	50,000.00



The numerical results demonstrate that the methods produce the same total depreciation when the same residual value is reached, but the timing of depreciation is materially different.

Interpretation of Results

- **Timing of expense** - Diminishing Balance recognizes substantially more expense in the early years. Its Year 1 depreciation of Rs 1,02,835.88 is more than twice the Straight-Line charge of Rs 45,000.
- **Book-value reduction** - At the end of Year 4, Straight-Line gives a book value of Rs 320,000, while Diminishing Balance gives approximately Rs 1,99,053.59. This difference is driven by accelerated early depreciation.
- **Total depreciation** - Both methods ultimately recognize Rs 4,50,000 because the asset starts at Rs 5,00,000 and is modelled to end at Rs 50,000. Therefore, the key distinction is timing rather than total depreciable cost.
- **Profit and reporting** - All else equal, Diminishing Balance would result in a larger depreciation expense and lower reported profit in earlier periods, followed by lower depreciation expense in later periods. Straight-Line creates a more stable annual expense profile. Actual tax treatment depends on the applicable jurisdiction and tax rules and is outside the supplied model.

Advanced Problem Statement

Sl. No.	Question	Answer
1	Calculate annual depreciation using Straight-Line.	45,000.00
2	Calculate total depreciation for the full life using Straight-Line.	4,50,000.00
3	Find depreciated book value after life using Straight-Line.	50,000.00
4	Calculate DB rate per year.	20.57%
5	Calculate depreciation in Year 2 using DB.	81,685.45
6	Find book value in Year 4 using DB.	1,99,053.59
7	Calculate total depreciation over full life using DB.	4,50,000.00
8	Find book value after life using DB.	50,000.00
9	Compare total depreciation under both methods.	0.00

Financial and Business

- **Consistency:** Straight-Line provides a predictable annual expense and is easy to incorporate into budgeting and forecasting.
- **Accelerated recognition:** Diminishing Balance is more reflective of assets expected to lose economic usefulness or value faster in earlier years.
- **Financial statement impact:** the choice changes the timing of expense and the carrying amount of the asset even though total depreciation remains the same under the model assumptions.
- **Method selection:** the method should reflect the expected pattern of economic benefit consumption and applicable accounting policy.
- **Decision usefulness:** comparing both schedules helps analysts understand how accounting assumptions can influence period-to-period profitability and asset values.

Advantages and Limitations

Method	Advantages	Limitation
Straight-Line	Simple, transparent and consistent; produces stable annual depreciation; useful when asset benefits are reasonably uniform.	May not represent assets whose economic value or productivity declines significantly faster in earlier years.
Diminishing Balance	Front-loads depreciation and can better represent assets with rapid early-life value loss, technological obsolescence or higher early-period consumption.	More complex and produces less uniform annual expense, which may reduce comparability of annual profit measures.

Actionable Recommendations

- Use Straight-Line when the asset is expected to provide relatively consistent benefits throughout its useful life.
- Consider Diminishing Balance when the asset is expected to lose value or economic usefulness more rapidly in the early years.
- Choose the method based on the asset's expected consumption pattern rather than simply seeking a preferred accounting outcome.
- For real-world accounting or tax reporting, validate the method against applicable accounting standards, tax regulations and company policy.

Model Controls and Limitations

This is an educational model based on the supplied assumptions. It does not incorporate tax-specific depreciation rules, component depreciation, impairment, revaluation, partial-year conventions, disposal proceeds or changes in useful-life and residual-value estimates. The Diminishing Balance schedule is calibrated to reach the specified Rs 50,000 scrap value at the end of Year 10.

The accompanying Excel model is formula-driven and includes inputs, methodology, schedules, advanced answers, dashboard/visual analysis and project documentation.

Conclusion

The analysis establishes that Straight-Line and Diminishing Balance can produce the same total depreciation over an asset's life when they use the same asset price and residual value, but they recognize that depreciation at different points in time. Straight-Line allocates Rs 45,000 every year, while Diminishing Balance recognizes a larger amount in the early years and progressively less later. For the project assumptions, Diminishing Balance produces Year 2 depreciation of Rs 81,685.45 and Year 4 ending book value of Rs 1,99,053.59, compared with Rs 45,000 annual depreciation and Rs 3,20,000 Year 4 ending book value under Straight-Line. Both methods end at Rs 50,000 and recognize Rs 4,50,000 total depreciation.

Overall insight: depreciation-method selection primarily changes the timing and pattern of expense recognition and book-value reduction. The appropriate method should be aligned with the asset's economic consumption pattern and the applicable accounting framework.